

# Recipes for Networking

It's always fun to try out different hacks under the GNU/Linux freedom platform. The pride of becoming a command-line wizard makes everyone stay close to the CLI. Moreover, the CLI vests you with the ultimate power to control your machine.

In this article, we'll explore networking under GNU/Linux. You'll find it interesting to manage the entire network through certain valid keystrokes known as commands. Imagine that you have to access the contents of several other machines from a mount point in your machine. Then imagine shutting down, rebooting, and installing applications on those remote machines, all at one time? Could you configure the WLAN and LAN interfaces from the CLI? This tutorial gives you some insights to the exciting bytes on controlling your network under GNU/Linux.

First we will learn the "Hello World" of a networked machine.

## Let us ping!

*ping* is a universal command that is available on every operating system to test the reachability of a network. When you shoot your terminal with the *ping* command and an IP address as its argument, the machine will try to send some bits of raw data towards the machine owning that IP address. If some machine exists with that IP address, it will send back certain bits. Thus the machine receives the bits and it confirms that a path is available from the current machine to the other through a network. We can check the existence of certain machines on the network by just *pinging*.

To see which machines are up in the current network, let's write a bit of shell script. Open *vim* as the root:

```
# vim /usr/bin/netup.sh
```

...and key in the following lines in it:

```
#!/bin/bash
for i in 192.168.1.{1..255}; // checks 192.168.1.x class of IPs.
do
ping -c 2 $i > /dev/null;
[ $? -eq 0 ] && echo $i is up.
```

```
trap "exit" SIGINT // To force exit when Ctrl+C keystroke is
applied.
done
```

Save the file, and make it executable by running the following command:

```
chmod a+x /usr/bin/netup.sh
```

Now, run the script as:

```
[slynu@gnuxbox ~]$ netup.sh
192.168.1.1 is up.
192.168.1.3 is up.
192.168.1.4 is up.
```

## Configuring your network

Now, let us look at how to configure your machine on the network. You can configure it using two methods. It can be configured manually by the *ifconfig* command for static IP addressing or it can be done via the DHCP (Direct Host Control Protocol).

Static IP addressing is the one that you explicitly instruct the system to use by giving an IP address for a given Ethernet or wireless interface. In case you're using the DHCP, simply issuing the *dhcclient* command will fetch the